

Memo

To: Court Auditor

Subject: Model Governance Review

Date: March 30, 2026

This memo summarizes findings of a COMPAS replacement risk model using logistic regression and gradient-boosted trees, along with explanation methods including SHAP, LIME, and counterfactual explanations. These methods were applied to 4 individuals, the highest and lowest-risk Black and White defendants in the dataset.

The explanation methods revealed that the model's predictions are primarily driven by prior criminal history, charge degree, and age category, with smaller contributions from demographic variables. Both SHAP and LIME indicated that individuals with a higher number of prior offenses and more serious charges were more likely to receive a high-risk prediction. In general, SHAP provided more stable global explanations across the dataset, while LIME provided more localized explanations for individual cases.

However, the explanation methods also have limitations. LIME explanations depend on local neighborhood sampling and can vary across runs, making them less stable for governance purposes. SHAP explanations are more consistent but can be computationally expensive as they require evaluating all possible feature coalitions. Counterfactual explanations were useful for showing what minimal changes would flip a prediction, but some counterfactuals may suggest changes that are unrealistic or involve immutable characteristics such as race or gender. These should not be interpreted as actionable recommendations.

The fairness analysis showed differences in error rates across racial groups, particularly in false positive and false negative rates. This indicates that model performance is not uniform across groups and should be monitored carefully. Explanation methods help understand model behavior but do not guarantee fairness or absence of bias.

I recommend ongoing monitoring of model performance by demographic group, periodic auditing of explanation outputs, and documentation of feature importance and counterfactual behavior. Immutable features such as race and gender should not be used in counterfactual recourse recommendations. Governance procedures should include regular fairness audits, model documentation, and human review for high-risk decisions to ensure the model is used responsibly and transparently.